

Part	Value	Device	Package	Description	Description2	Qty	Place_YE S/NO	Provided_b y_custom r_YES/NO	Distributor	Ordernumber	Remarks_Customer
U9,U7,U5	LMG2100R044RARR	LMG2100R044RARR	VQFN-FCRLF-16	100-V 4.4-mΩ half-bridge GaN FET with integrated driver and protection		3	YES	NO	Mouser	595-LMG2100R044RARR	
C78,C5,C17,C14,C15,C16	10uF	GRM32EC72A106KE05L	C_1210	Murata, GRM, Oberfläche Keramik- Vielschichtkondensator Keramik, 10 µF 10 % / 100V dc		6	YES	NO	RS Components	247-8166	
C10,C8,C31,C30,C60,C45,C44,C61,C59,C46,C29,C6	1uF	CGA4J3X7S2A105K125AE	C_0805	TDK, CGA, Oberfläche Keramik- Vielschichtkondensator X7S, 1 µF ±10 % / 100V dc, Gehäuse 0805		12	YES	NO	RS Components	915-5263	
J7,J1	Wurth-634008	634008149821	Wurth_634008_RJ45	Buchse Cat.5 RJ45- Steckverbinder, 1-Port 8-polig Messing vernickelt Oberfläche Horizontal		2	YES	NO	RS Components	162-2309	
C49,C35,C36,C53,C50,C7,C4,C65,C1,C55,C2,C3,C79,C68,C38,C51,C34,C40,C66,C9,C25,C64,C76,C77,C21,C20	100nF	VJ0603Y104KXBCW1B C	C_0603	Vishay, W1BC, Oberfläche Keramik- Vielschichtkondensator X7R, 0.1 µF ±0.1 % / 100V dc, Gehäuse 0603		26	YES	NO	RS Components	272-7888P	
R11,R25,R18	3m	BVT-I-R003-1.0	R_2512	Isabellenhütte BVT, AEC-Q200 Zeranin SMD- Widerstand 3 mΩ ±1 % / 3 W 50 ppm/K		3	YES	NO	RS Components	193-0036	
R14,R21,R10,R17,R24,R7	10R	CRCW060310R0FKEA	R_0603	Vishay CRCW, AEC-Q200 Dickschicht SMD-Widerstand 10 Ω ±1 % / 0.1 W -100 ppm/°C, +100 ppm/°C		6	YES	NO	RS Components	678-9695	
D4,D3	LED	LS R976-NR-1	LED_0805	ams OSRAM CHIP LED 0805 Oberfläche LED Rot 2 V, 160 °, 2-Pin		2	YES	NO	RS Components	497-4804	
C33,C47,C63,C62,C48,C32	47pF	C0603C470J5GACTU	C_0603	KEMET, C, Oberfläche Keramik- Vielschichtkondensator C0G, 47 pF ±5 % / 50V dc		6	YES	NO	RS Components	264-4517	
U10,U8,U6	INA240A1PW	INA240A1PWR	TSSOP-8_4.4x3mm_P0.65mm	Texas Instruments Stromsensor 25 µV 1.8 mA TSSOP, 8-Pin Oberfläche		3	YES	NO	RS Components	226-4772	
C57,C43,C42,C58,C26,C41,C27,C28,C56	220nF	GCM188R71H224KA64J	C_0603	Murata, GCM, Oberfläche Keramik- Vielschichtkondensator X7R, 220 nF ±10 % / 50V dc		9	YES	NO	RS Components	247-7879P	
R16,R22,R15,R29,R9,R8,R23,R33,R35,R36,R32,R34,R31	20R	RS Best.-Nr.: 243-0578P	R_0603	RS PRO RSTN Dünnschicht Widerstand 22 Ω 0.5 % / 0.15 W ±50 to 100 ppm/°C		13	YES	NO	RS Components	243-0578P	20Ω not available; 22Ω as alternative

R46,R45	2.2k	RS Best.-Nr.: 243-0666	R_0805	RS PRO RSTN Dünnschicht Widerstand 2.2 kΩ 0.1 % / 0.4 W ±50 to 100 ppm/°C		2	YES	NO	RS Components	243-0666	
Q1	IPB020N10N5	IPB020N10N5ATMA1	TO-263-2	Infineon IPB020N10N5 Typ N- Kanal, Oberfläche MOSFET Erweiterung 100 V / 176 A 375 W		1	YES	NO	RS Components	171-1955	
R1,R3,R19,R12,R5,R26	100k	RT0603BRD07100KL	R_0603	Yageo RT Dünnschicht SMD- Widerstand 100 kΩ ±0.1 % / 0.1 W 25 ppm/°C		6	YES	NO	RS Components	199-1385	
U2	ISM330BXTR	ISM330BXTR	LGA- 14_3x2.5mm_P0.5mm_LayoutBo rder3x4y	STMicroelectronics Näherungsschalter 6-Achsen Oberfläche 1.71 V 3.6V 2.8 kHz Trägheitssensor		1	YES	NO	RS Components	151-685	
R37,R4,R39,R38,R2,R28, R30	33R	CRCW060333R0FKEA HP	R_0603	Vishay CRCW, AEC-Q200 Dickschicht SMD-Widerstand 33 Ω ±1 % Oberfläche / 0.25 W -100 ppm/K, +100 ppm/K		7	YES	NO	RS Components	820-6837	
C69,C75,C73,C70,C72,C7 1,C74	2.2nF	CGA3E2C0G1H222J08 0AA	C_0603	TDK, CGA, Oberfläche Keramik- Vielschichtkondensator C0G, 2.2 nF ±5 % / 50V dc		7	YES	NO	RS Components	915-5705P	
C24,C54,C39	2.2uF	C2012X7R1H225K125 AC	C_0805	TDK, C, Oberfläche Keramik- Vielschichtkondensator X7R, 2.2 µF ±10 % / 50V dc		3	YES	NO	RS Components	788-3048P	
C80	47nF	C0603C473K1RACTU	C_0603	KEMET, C, Oberfläche Keramik- Vielschichtkondensator X7R, 47 nF ±10 % / 100V dc		1	YES	NO	RS Components	801-5384	
J5	RS PRO 528 D-Sub 2.77mm	RS Best.-Nr.: 239-5855	DSUB- 9_Female_Horizontal_P2.77x2.84 mm_EdgePinOffset4.94mm_Hou sed_MountingHolesOffset7.48m m	RS PRO D-Sub- Steckverbinder D gewinkelt, 9- polig 2.77 mm Raster, Durchsteckmontage Buchse		1	YES	NO	RS Components	239-5855	
C12,C13	470uF	ECA2AHG471	CP_Radial_D16.0mm	Panasonic NHG Radial, Durchsteckmontage Kondensator 470 µF Radiale Anschlussfahnen ±20 % / 100V dc		2	YES	NO	RS Components	365-4543P	
C11	47uF	GRM32EC81C476KE15 L	C_1210	Murata, GRM, Oberfläche Keramik- Vielschichtkondensator X6S, 47 µF ±10 % / 16V dc		1	YES	NO	RS Components	846-7369	
L1	22uH	74438356220	L_Wuerth_MAPI-4020	Würth Elektronik WE-MAPI SMT Power-Induktivität mit Abschirmung, 22 µH 1.85 A AEC-Q200 mit Mangan-Zink- Kern		1	YES	NO	RS Components	258-5689	

U3	L6982C50DR	L6982C50DR	SO-8_3.9x4.9mm_P1.27mm	STMicroelectronics Spannungsregler 38V, 8-Pin		1	YES	NO	RS Components	261-4749P	
J2	Würth Elektronik 2365 Printklemme	691236510003	Würth Elektronik 2365 Printklemme	Würth Elektronik 2365 Printklemme Horizontal, Durchsteckmontage, 3-polig / 1-reihig, Raster 5.08 mm		1	YES	NO	RS Components	191-0204	
U13	LTC4359	LTC4359IS8#PBF	SO-8_3.9x4.9mm_P1.27mm	Power Management Specialised - PMIC Ideal Diode Cntr w/ Reverse In Protectopm		1	YES	NO	Mouser	584- LTC4359IS8#P BF	
U1	AS5047P-ATSM	AS5047P-ATSM	TSSOP-14_4.4x5mm_P0.65mm	ams OSRAM 14-Bit- Magnetischer Drehpositionssensor auf der Achse 1-Achsen Oberfläche 3 V 7V SPI		1	YES	NO	RS Components	200-9464	
C67,C52,C37,C23	33nF	C0603C333K5RACTU	C_0603	KEMET, C, Oberfläche Keramik- Vielschichtkondensator X7R, 33 nF ±10 % / 50V dc		4	YES	NO	RS Components	147-869	
R41,R48,R40,R47	0R	MCR03SEQPJ000	R_0603	ROHM MCRC Series, AEC- Q200 Dickschicht Widerstand 0 Ω ±5 % Oberfläche / 0.125 W ±100 ppm/°C		4	YES	NO	RS Components	780-542	
C22,C18,C19	4.7uF	C0805C475K3RACAUT O	C_0805	KEMET, VW 80828 Keramik- Vielschichtkondensator X7R, 4.7 µF 10 % / 50V		3	YES	NO	RS Components	188-2142P	
U4	AP2114D	AP2114D-3.3TRG1	TO-252-2	DiodesZetex Regler, Rauscharmer LDO 5 1 A TO- 252 3.3V		1	YES	NO	RS Components	828-3515	
R27,R6,R20,R13	4.22k	CRCW06034K22FKEA	R_0603	Vishay CRCW, AEC-Q200 Dickschicht SMD-Widerstand 4.22 kΩ ±1 % / 0.1 W -100 ppm/°C, +100 ppm/°C		4	YES	NO	RS Components	679-0465	
U12	MCP2562FD-E/MF	MCP2562FD-E/MF	DFN-8- 1EP_3x3mm_P0.65mm_EP1.55x 2.4mm	Microchip CAN-Transceiver, 8 Mbit/s 1 Transceiver Standby 70 mA		1	YES	NO	RS Components	824-3135	
R42	120R	RN73R1JTDD1200B25	R_0603	KOA RN73R, AEC-Q200 Metallschicht SMD- Widerstand 120 Ω ±0.1 % / 0.1 W ±25 ppm/°C		1	YES	NO	RS Components	253-3100	

U11	NUCLEO-G431RB	ESQ-119-24-T-D	2x19 Leiterplattenbuchse; 2.54mm Raster	Samtec ESQ Leiterplattenbuchse Gerade Durchsteckmontage 38-polig / 2-reihig, Raster 2.54 mm Lot		2	NO	NO	RS Components	196-9830	Two articles are needed to fill a single footprint; Please Also Order
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J4,J3	RS Pro 304-02-818	RS Best.-Nr.: 187-3802	S16N-PC 4mm Shrouded socket	RS PRO 4 mm Buchse Schwarz, Rot Buchse für Laborstecker, Kontakt Gold, 1kV 24A Ummantelte Buchse		1	NO	NO	RS Components	187-3802	A single article fills two footprints; Please Also Order
J6	Conn_02x03_Pin	TSW-106-08-L-D-RA	PinHeader_2x03_P2.54mm_Hori zontal	Samtec TSW Stiftleiste gewinkelt, 12-polig / 2-reihig, Raster 2.54 mm Raster Durchsteckmontage		1	NO	NO	RS Components	767-0934	Article has 12 poles; Cut to fit; Please Also Order
-	9471LE (1/2 IN X 60 YD)	-	-	hermal Interface Products Adhesive Thermal Transfer Tape, Clear, 1/2 in x 60 yd, 2 mil		1	NO	NO	Mouser	517-9471LE- 1/2IN	Non-electrical item; Please Also Order
-	Fischer Elektronik Kühlkörper L	ICK S 40x40x20	-	Fischer Elektronik Kühlkörper für Universelle quadratische Alu, 40 mm x 40mm x 20 mm		1	NO	NO	RS Components	674-4835	Non-electrical item; Please Also Order
-	Fischer Elektronik Kühlkörper S	ICK SMD F 8MI	-	Fischer Elektronik ICKSMDF8SA Kühlkörper für Universelle rechteckige Alu, 8 mm x 8mm x 6 mm		3	NO	NO	RS Components	692-0694	Non-electrical item; Please Also Order